$$B = \frac{a_{11}}{x_1x_1} + \frac{a_{12}}{x_1x_2} + \frac{a_{13}}{x_1x_3} + \frac{a_{14}}{x_1x_4} + \frac{a_{22}}{x_2x_2} + \frac{a_{23}}{x_2x_3} + \frac{a_{24}}{x_2x_4} + \frac{a_{33}}{x_3x_3} + \frac{a_{34}}{x_3x_4} + \frac{a_{44}}{x_4x_4}$$

$$+ \frac{a_{00}}{x_0^2} + \frac{a_{01}}{x_0x_1} + \frac{a_{02}}{x_0x_2} + \frac{a_{03}}{x_0x_3} + \frac{a_{04}}{x_0x_4} = \sum_{i \le j} \frac{a_{ij}}{x_i x_j}$$

the possible constraints are

$x_0 = X_{35}$

$x_1 = X_{13}$ $\quad x_2 = X_{14}$

$x_3 = X_{24}$ $\quad x_4 = X_{25}$

① $x_4 = -x_1$ $\quad x_0$ independent $\quad \leftarrow c_{13} = c_{14} = 0$

$x_3 = x_2 - x_1$

since x_0 is arbitrary, all the terms with $\frac{1}{x_0^2}$ vanish, also all terms with $\frac{1}{x_0}$

- $\frac{a_{00}}{x_0^2} = 0 \quad \forall x_0 \Rightarrow \boxed{a_{00} = 0}$

- $\frac{1}{x_0}\left(\frac{a_{01}}{x_1} + \frac{a_{02}}{x_2} + \frac{a_{03}}{x_3} + \frac{a_{04}}{x_4}\right) = 0 \quad \forall x_0 \Rightarrow \frac{a_{01}}{x_1} + \frac{a_{02}}{x_2} + \frac{a_{03}}{x_3} + \frac{a_{04}}{x_4} = 0$

(when $\begin{cases} x_4 = -x_1 \\ x_3 = x_2 - x_1 \end{cases}$)

fix x_2, let $x_1 \to \infty$ $\quad (|x_3|, |x_4| \to \infty) \quad \to 0 = \frac{a_{02}}{x_2}$ $\quad$ so $\boxed{a_{02} = 0}$

fix x_3 let $x_1 \to \infty$ $\quad (|x_2|, |x_4| \to \infty) \quad \to 0 = \frac{a_{03}}{x_3}$ $\quad$ so $\boxed{a_{03} = 0}$

so now we have $\frac{a_{01}}{x_1} + \frac{a_{04}}{x_4} = 0$ when $x_4 = -x_1$ $\quad$ so $\boxed{a_{01} = a_{04}}$

$\rightarrow$ look like $n=4$

---

② $x_0 = -x_2$ $\quad x_1$ independent

$x_4 = x_3 - x_2$

$\frac{a_{11}}{x_1^2} = 0 \qquad \boxed{a_{11} = 0}$

$\frac{1}{x_1}\left(\frac{a_{10}}{x_0} + \frac{a_{12}}{x_2} + \cancel{\frac{a_{13}}{x_3}} + \cancel{\frac{a_{14}}{x_4}}\right) = 0 \quad \rightarrow \boxed{a_{10} = a_{12}}$

---

③ $x_1 = -x_3$ $\quad x_2$ independent

$x_0 = x_4 - x_3$

$\frac{a_{22}}{x_2^2} = 0 \quad \rightarrow \boxed{a_{22} = 0}$

$\frac{1}{x_2}\left(\cancel{\frac{a_{20}}{x_0}} + \frac{a_{21}}{x_1} + \frac{a_{23}}{x_3} + \cancel{\frac{a_{24}}{x_4}}\right) \rightarrow \boxed{a_{21} = a_{23}}$

(4) $x_2 = -x_4$ $\quad$ x_3 independent

$x_1 = x_0 - x_4$

$a_{33} = 0$

$$\cancel{\frac{a_{30}}{x_0}} + \cancel{\frac{a_{31}}{x_1}} + \frac{a_{32}}{x_2} + \frac{a_{34}}{x_4} = 0 \qquad \boxed{a_{32} = a_{34}}$$

---

(5) $x_3 = -x_0$ $\quad$ x_4 independent

$x_2 = x_1 - x_0$

$\boxed{a_{44} = 0}$

$$\frac{a_{40}}{x_0} + \cancel{\frac{a_{41}}{x_1}} + \cancel{\frac{a_{42}}{x_2}} + \frac{a_{43}}{x_3} = 0 \qquad \boxed{a_{40} = a_{43}}$$

$$B = \cancel{\frac{a_{11}}{x_1 x_1}} + \frac{a}{x_1 x_2} + \cancel{\frac{a_{13}}{x_1 x_3}} + \cancel{\frac{a_{14}}{x_1 x_4}} + \cancel{\frac{a_{22}}{x_2 x_2}} + \frac{a}{x_2 x_3} + \cancel{\frac{a_{24}}{x_2 x_4}} + \cancel{\frac{a_{33}}{x_3 x_3}} + \frac{a}{x_3 x_4} + \cancel{\frac{a_{44}}{x_4 x_4}}$$

$$+ \cancel{\frac{a_{00}}{x_0^2}} + \frac{a}{x_0 x_1} + \cancel{\frac{a_{02}}{x_0 x_2}} + \cancel{\frac{a_{03}}{x_0 x_3}} + \frac{a}{x_0 x_4}$$

---

why are these not all non-bcd terms?

$n = 6$ $\quad$ 9 variables

$\boxed{x_{13}\ x_{14}\ x_{15}}$

$\boxed{x_{24}\ x_{25}\ x_{26}}$

$\boxed{x_{35}\ x_{36}}$

$\boxed{x_{46}}$

Gutenit (1)

$x_{24} = x_{14} - x_{13}$

$x_{25} = x_{15} - x_{13}$

$x_{26} = -x_{13}$

($x_{35}\ x_{36}\ x_{46}$ indep)

$c_{13} = c_{14} = c_{15} = 0$

cyclic $\rightarrow$ (2)

$x_{35} = x_{25} - x_{24}$

$x_{36} = x_{26} - x_{24}$

$x_{31} = -x_{24}$

($x_{46}\ x_{41}\ x_{51}$)

$c_{24} = c_{25} = c_{26} = 0$

previous way of naming them

$\rightarrow$ wanted Gutenit to be

(1)

$x_4 = x_2 - x_1$

$x_5 = x_3 - x_1$

$x_7 = -x_1$

$x_1 = x_{13}$ $\quad$ $x_4 = x_{24}$

$x_2 = x_{14}$ $\quad$ $x_5 = x_{25}$

$x_3 = x_{15}$

$x_7 = ?$ $\quad$ $x_8 = ?$ $\quad$ $x_9 = ?$

---

(2)

$? = x_5 - x_4$

$? = x_7 - x_4$

$x_1 = -x_4$

$13 \rightarrow 24 \rightarrow 35 \rightarrow 46 \rightarrow 51 \rightarrow 62 \rightarrow 13$

$x_1 \quad x_2 \quad x_3 \quad x_4 \quad x_5 \quad x_6$

$14 \rightarrow 25 \rightarrow 36$

$y_1 \quad y_2 \quad y_3$

$x_{24} = x_{14} - x_{13}$

$x_{25} = x_{15} - x_{13}$

$x_{26} = -x_{13}$

$$\boxed{\begin{aligned} x_2 &= y_1 - x_1 \\ y_2 &= x_5 - x_1 \\ x_6 &= -x_1 \end{aligned}} \xrightarrow{+1} \boxed{\begin{aligned} x_3 &= y_2 - x_2 \\ y_3 &= x_6 - x_2 \\ x_1 &= -x_2 \end{aligned}}$$

$x_{35} = x_{25} - x_{24}$

$x_{36} = x_{26} - x_{24}$

$x_{31} = -x_{24}$

indep: $x_3 \; x_4 \; y_3$

indep $x_4 \; x_5 \; y_1$

$n = 7$

$13 \to 24 \to 35 \to 46 \to 57 \to 61 \to 72$

$x_1 \quad x_2 \quad x_3 \quad x_4 \quad x_5 \quad x_6 \quad x_7$

$14 \to 25 \to 36 \to 47 \to 51 \to 62 \to 73$

$y_1 \quad y_2 \quad y_3 \quad y_4 \quad y_5 \quad y_6$

$\frac{7}{2}$ 4

$$\boxed{\begin{aligned} x_2 &= y_1 - x_1 \\ y_2 &= y_5 - x_1 \end{aligned}}$$

$\to y_6 = x_6 - x_1$ $\to x_{n-1}$

$\boxed{x_7 = -x_1}$ $\to x_n$

$13 \to 24 \to 35 \to 46 \to 57 \to 68 \to 71 \to 82$

$x_1 \quad x_2 \quad x_3 \quad x_4 \quad x_5 \quad x_6 \quad x_7 \quad x_8$

$14 \to 25 \to 36 \to 47 \to 58 \to 61 \to 72 \to 83$

$y_1 \quad y_2 \quad y_3 \quad y_4 \quad y_5 \quad y_6 \quad y_7 \quad y_8$

$15 \to 26 \to 37 \to 48$

---

try $n = 6$

\# variables : $\frac{n(n-3)}{2} = 9$

\# constrained variables $= 2(n-3) = 6$

\# independent $= 3$

$$B = \sum_{k=1}^{9} \left[\underbrace{\frac{a_{kkk}}{x_k^3}}_{\to 0} + \frac{1}{x_k^2} \underbrace{\left(\sum_{i \neq k} \frac{a_{kki}}{x_i} \right)}_{\text{boxs line } n=4} + \frac{1}{x_k} \underbrace{\left(\sum_{\substack{j>i \\ i,j \neq k}} \frac{a_{kij}}{x_i x_j} \right)}_{\text{boxs line } n=5} + \sum_{\substack{i<j<l \\ i,j,l \neq k}} \frac{a_{ijl}}{x_i x_j x_l} \right]$$

for each k there is one constraint in which x_k independent

$$\to \sum_{i \neq k} \frac{a_{kki}}{x_i} = 0 \;, \quad \sum_{\substack{j>i \\ i,j \neq k}} \frac{a_{kij}}{x_i x_j} = 0 \;, \quad \frac{a_{kkk}}{x_k^3} = 0$$

$\boxed{k=1}$

① $\boxed{a_{111}=0}$

② $\displaystyle\sum_{i=2}^{9} \frac{a_{11i}}{x_i} = 0$

$x_5 = x_3 - x_2$
$x_6 = x_4 - x_2$
$x_7 = -x_2$

- $x_2 \to \infty$, x_3, x_4 fixed $\to$ $x_7, x_5, x_6 \to -\infty$ $\quad\|\quad$ survive: $0 = \dfrac{a_{113}}{x_3} + \dfrac{a_{114}}{x_4} \quad \forall\, x_3, x_4$

$\to \boxed{a_{113} = a_{114} = 0}$

- $x_7 \to \infty$, x_5, x_6 fixed $\to$ $x_2, x_3, x_4 \to \infty$ $\quad\|\quad$ $0 = \dfrac{a_{115}}{x_5} + \dfrac{a_{116}}{x_6} = 0 \quad \forall\, x_5, x_6$

$\to \boxed{a_{115} = a_{116} = 0}$

left with $\dfrac{a_{112}}{x_2} + \dfrac{a_{117}}{x_7} + \dfrac{a_{118}}{x_8} + \dfrac{a_{119}}{x_9} = 0$

x_8, x_9 are independent so $\boxed{a_{118} = a_{119} = 0}$

$\dfrac{a_{112}}{x_2} + \dfrac{a_{117}}{x_7} = 0$ but $x_7 = -x_2$ $\to$ $\boxed{a_{112} = a_{117}}$

$\to$ arbitrary $n=4$ amplitude

---

next $\boxed{\displaystyle\sum_{j \geq i > 1} \frac{a_{1ij}}{x_i x_j} = 0}$

36 terms

want 15 $\to$ need to get rid of 21

x_8, x_9 independent $\to$ $a_{188} = a_{199} = a_{189}$ ③

$\displaystyle\sum_{i \neq 1,8,9} \frac{a_{1i8}}{x_i} = 0$ $\to$ looks like $n=4$ ⑥

$\displaystyle\sum_{i \neq 1,8,9} \frac{a_{1i9}}{x_i} = 0$ ⑥

got rid of 15, need 6 more

limits

$x_5 = x_3 - x_2$

$x_6 = x_4 - x_2$

$x_7 = -x_2$

$$\sum_{\substack{j \geq i > 1 \\ i,j \neq 8,9}} \frac{a_{ij}}{x_i x_j} = 0$$

- $x_2 \to \infty$, x_3, x_4 fixed $\to$ $x_7, x_5, x_6 \to -\infty$

$$\frac{a_{133}}{x_3^2} + \frac{a_{134}}{x_3 x_4} + \frac{a_{144}}{x_4^2} = 0 \qquad \forall\, x_3, x_4$$

- $x_7 \to \infty$, x_5, x_6 fixed $\to$ $x_2, x_3, x_4 \to \infty$

$$\frac{a_{155}}{x_5^2} + \frac{a_{156}}{x_5 x_6} + \frac{a_{166}}{x_6^2} = 0 \qquad \forall\, x_5\, x_6$$

$\boxed{\text{we're left with } 15}$

Same # as case $n=5$

remind: i or $j \in \{8,9\}$ $\longrightarrow$ $(28)(38)(48)(58)(68)(78)(88)(89)$
$(29)(39)(49)(59)(69)(79)(99)$

$ij = (33), (34), (44), (55), (56), (66)$

22	23	—	28	29	8
	33	—	—	39	7
			44	49	6
				99	1

$2 \cdot 8 - 1 = 15$

$$\frac{a_{122}}{x_2^2} + \frac{a_{123}}{x_2 x_3} + \frac{a_{124}}{x_2 x_4} + \frac{a_{125}}{x_2 x_5} + \frac{a_{126}}{x_2 x_6} - \frac{a_{127}}{x_2^2}$$

$$\frac{a_{135}}{x_3 x_5} + \frac{a_{136}}{x_3 x_6} - \frac{a_{137}}{x_3 x_2}$$

$$\frac{a_{145}}{x_5 x_6} + \frac{a_{146}}{x_4 x_6} - \frac{a_{147}}{x_4 2}$$

$$-\frac{a_{157}}{x_5 x_2}$$

$$-\frac{a_{167}}{x_6 x_2}$$

$$\frac{a_{177}}{x_2^2}$$

$x_5 = x_3 - x_2$

$x_6 = x_4 - x_2$

$\longrightarrow$ $n = 5$ case

$x_2 = x_4 - x_6$

$x_5 = x_3 - x_7$

$x_6 = x_4 - x_7$

$x_2 = -x_7$

by redoing step 1 6 times

$e_{kkk} = 0 \quad \forall k$

x_{35} x_{36} x_{46} free variables

shift 6 times

- x_{46} x_{41} $\boxed{x_{51}}$
- $\boxed{x_{51}}$ $\boxed{x_{52}}$ $\boxed{x_{62}}$
- $\boxed{x_{62}}$ x_{63} x_{13}
- x_{13} x_{14} $\boxed{x_{24}}$
- $\boxed{x_{24}}$ $\boxed{x_{25}}$ x_{35}

each free variable appears in 2 1-zeros